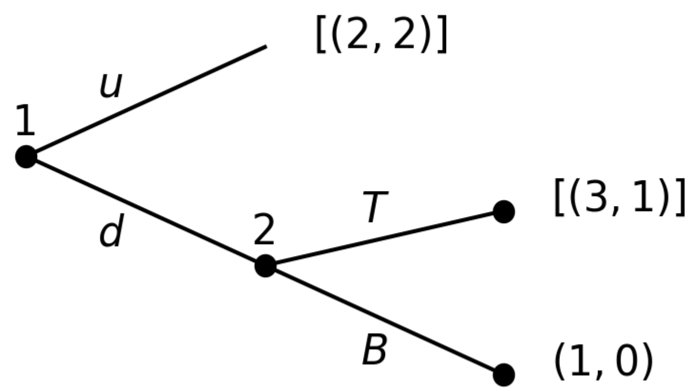
Imperfect information

Refinements—Imperfect information

- Refinements of extensive form games
- Weakly Perfect Bayesian Equilibrium (WPBE):
- Sequential Equilibrium (SE)
- Forward induction concepts.

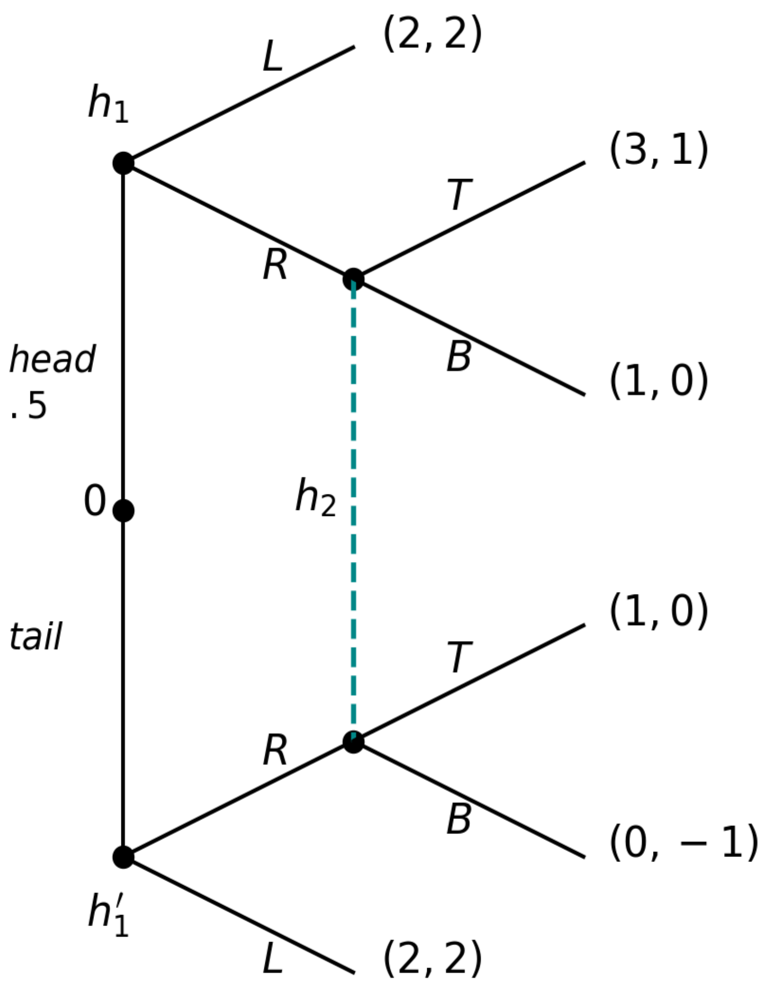
Non-credible threats, perfect information



- Two Nash equilibria
  - (d, T).
  - (u, B). Not SPE. Subgame perfection eliminates noncredible threats in games of perfect information.

Threats and imperfect information

Example 1



NE: $[(L, L), B]$, $(EU_1, EU_2) = (2, 2)$

NE: $[(R, L), T]$, $(EU_1, EU_2) = (2.5, 1.5)$

$[(L, L), B]$ is an unintuitive NE

- This is a signaling game:
  - Nature flips a coin.
  - Player 1 observes either h_1 (head) or h'_1 (tail), and chooses an action.
  - Player 2 does not observe nature's move but observes player 1's action and responds.
- The game has no subgames, so every NE is SPE.
- Equilibrium $[(L, L, B)]$ is *not* intuitive.
  - For player 2, B is a dominated strategy, and therefore it should not be credible. Subgame perfection does not apply.

WPBE

Assessment

Fix a finite extensive-form game Γ.

Definition 1 (Assessment)

- A system of beliefs is a profile $\mu = (\mu_1, \ldots, \mu_I)$ such that

$$(\forall i \in N \setminus \{0\})\ (\forall h \in \mathcal{U}^i) \quad \mu_i(\cdot|h) \in \Delta(h),$$

  where $\Delta(h)$ is the set of probability distributions on the information set h.

For $x \in h$, $\mu_i(x|h)$ is the probability assigned to x by beliefs $\mu_i(\cdot|h)$.

An assessment is a strategy-profile and system-of-beliefs pair (σ, μ).

Weakly Perfect Bayesian Equilibrium

Definition 2 (WPBE) An assessment (σ, μ) is a Weakly Perfect Bayesian Equilibrium (WPBE) of a finite, extensive-form game if

1. Sequential rationality: $(\forall i \in N)\ (\forall h \in \mathcal{U}_i)\ (\forall \sigma_i' \in \Sigma_i)$

$$E[u_i | h, \mu_i, \sigma_i, \sigma_{-i}] \geq E[u_i | h, \mu_i, \sigma_i', \sigma_{-i}]$$

2. Bayes rule: $(\forall i \in N)(\forall h \in \mathcal{U}_i)$,

$$P_\sigma(h) > 0 \implies \forall x \in h,\ \mu_i(x|h) = \frac{P_\sigma(x)}{P_\sigma(h)}$$

NE and sequential rationality

Fix a finite, extensive-form game Γ of perfect recall.

Theorem 1 Let (σ, μ) be an assessment. If at every information set in the path of play,

i. μ is derived by Bayes rule from σ and

ii. (σ, μ) satisfies sequential rationality, then σ is a NE of Γ.

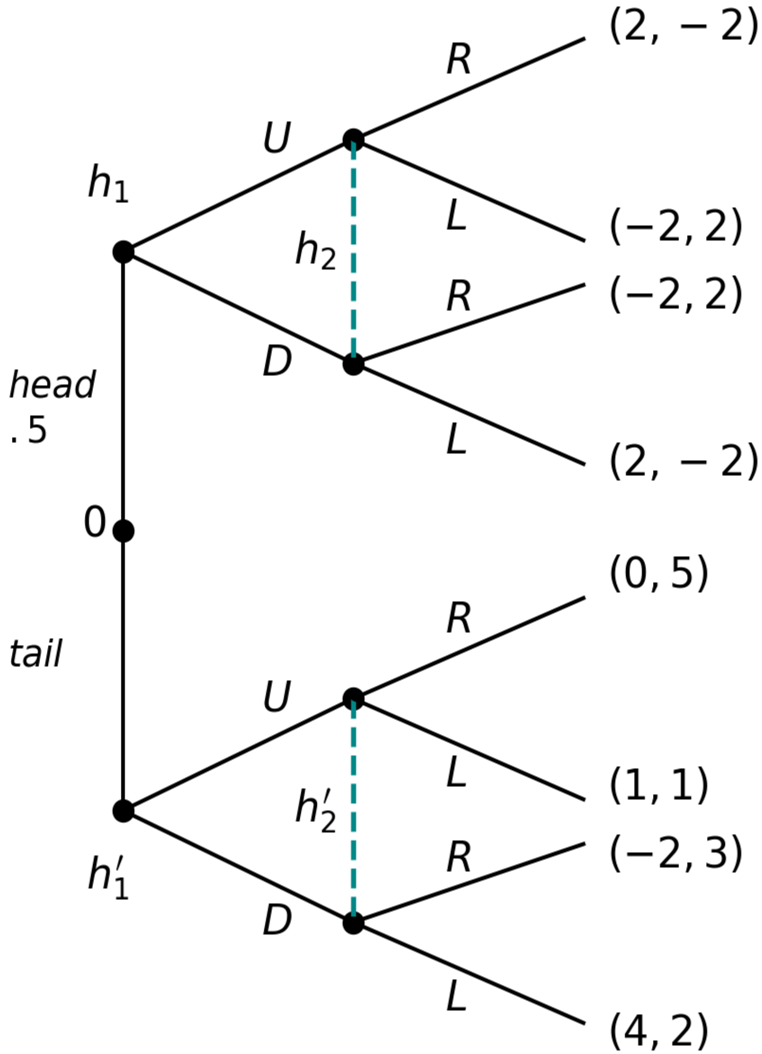
Theorem 2 Let σ be a NE and μ a system of beliefs derived by Bayes rule from σ when possible. Then (σ, μ) is sequentially rational at every information set on the equilibrium path.

- Maschler, Solan, and Zamir (2013), Theorems 7.44, 7.45. pp. 276-277.

Sequential rationality. Interpretation

- Subgame perfection eliminates non-credible threats by requiring that a NE be played in any subgame.
- Games of imperfect information may have no subgames. A ``game” starting at any information set may be defined.
- Interpret an information set h as the beginning of the game; $\mu_i(\cdot|h)$ as the probability of being at each $x \in h$.
- Available actions remain as in the original game.
- $E[u_i|h, \mu_i, \sigma_i, \sigma_{-i}]$ denotes i’s expected payoff in the ``game” that begins at information set h.

Example to illustrate WPBE 1



NE: $[(L, L), B]$, $(EU_1, EU_2) = (2, 2)$

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$[(L, L), B]$ is an unintuitive NE

Player 1's strategy and beliefs 2

$$\sigma_1(h_1) = \underbrace{(0.5,0.5)}_{(U,D)}$$

$$\sigma_1(h_1') = \overbrace{(1,0)}$$

$$\mu_1(head|h_1) = 1$$

$$\mu_1(tail|h_1') = 1$$

Player 2's strategy and beliefs 3

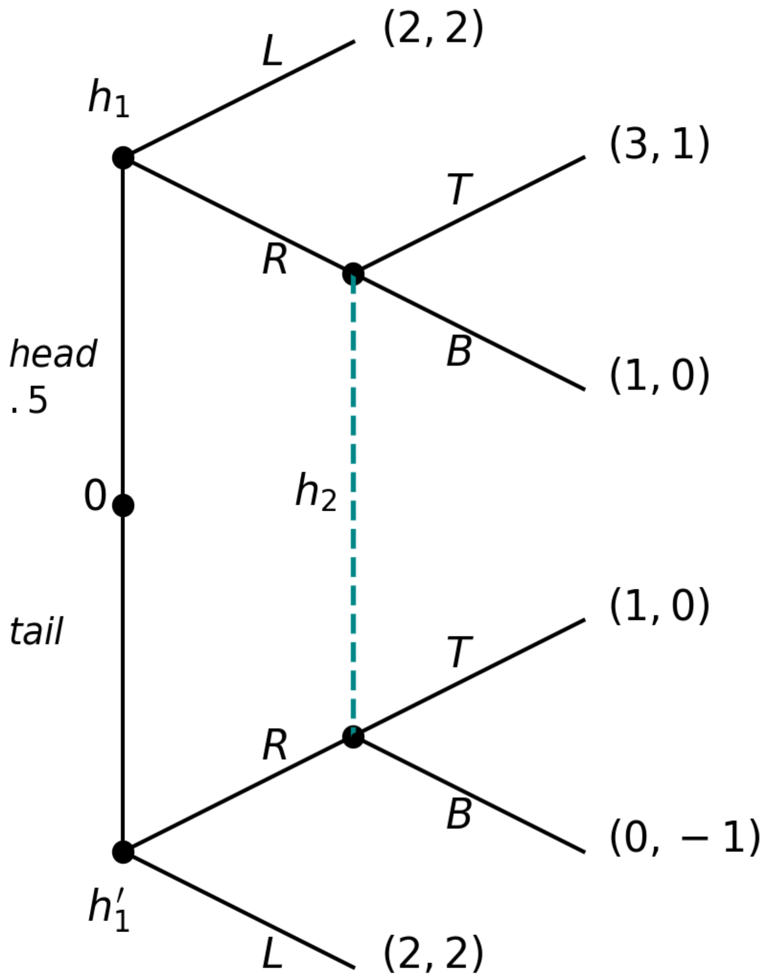
$$\sigma_2(h_2) = \underbrace{(0.5,0.5)}_{(R,L)}$$

$$\sigma_2(h_2') = \overbrace{(1,0)}$$

Consider the beliefs:

$$\mu_2((head, U)|h_2) = 0.5$$
$$\mu_2((tail, U)|h_2') = 1$$

Revisit Example 1



NE: $[(R, L), T]$, $\quad (EU_1, EU_2) = (2.5, 1.5)$

1's beliefs: $\mu_1(head|h_1) = 1,\ \mu_1(tail|h_1')) = 1$

2's beliefs: $\mu_2((head, R)|h_2) = 1$

- 2's beliefs: $h_2 = \{(head, R), (tail, R)\}$.

$$
\begin{aligned}
&\mu_2((head, R)|h_2) \\
&\quad = \frac{P_\sigma((head, R))}{P_\sigma(h_2)} \\
&\quad = \frac{P_0(head) \times \sigma_1(R|h_1)}{P_0(head) \times \sigma_1(R|h_1) + P_0(tail) \times \sigma_1(R|h_1')} \\
&\quad = \frac{1/2 \times 1}{1/2 \times 1 + 1/2 \times 0} = \frac{1/2}{1/2} = 1
\end{aligned}
$$

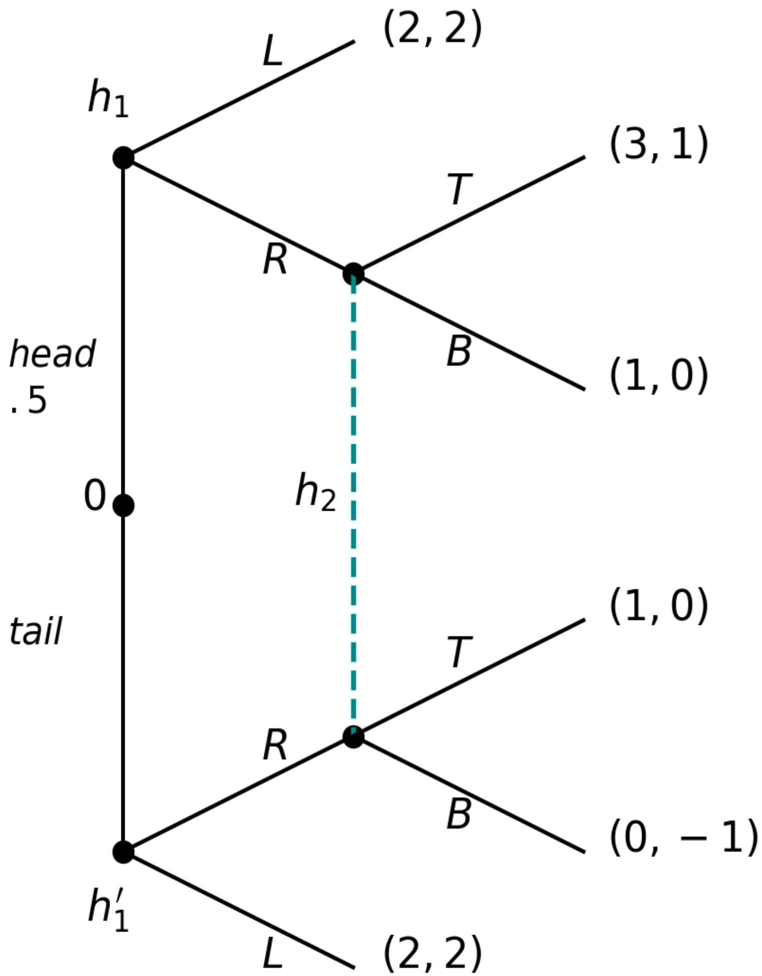
- 2's sequential rationality follows. See previous figure.

- 2’s sequential rationality if not yet convinced.

$$E[U_2|h_2, T, \sigma_1] = \mu_2((head, R)|h_2)1 + \mu_2((tail, R)|h_2)0 = 1$$
$$E[U_2|h_2, B, \sigma_1] = \mu_2((head, R)|h_2)0 + \mu_2((tail, R)|h_2)(-1) = 0$$

$$E[U_2|h_2, T, \sigma_1] = 1 > 0 = E[U_2|h_2, B, \sigma_1]$$

The other equilibrium in Example 1



NE: $[(L, L), B]$, $\quad (EU_1, EU_2) = (2, 2)$.

$[(L, L), B]$ is the unintuitive NE.

$P_\sigma(h_2) = 0$; then free to choose beliefs.

$(\forall \mu_2(x|h_2) \in [0, 1])$ 2's best action is T.

No beliefs support 2's choosing B at h_2.

Hence $[(L, L), B]$ is not a WPBE.

Example: WPBE need not be SPE

- A potential entrant (E), decides whether to enter a market, (in) or (out).
- If E chooses in, then E and the incumbent (I), decide simultaneous whether to fight or acquiesce. Incumbent actions are in lower case.

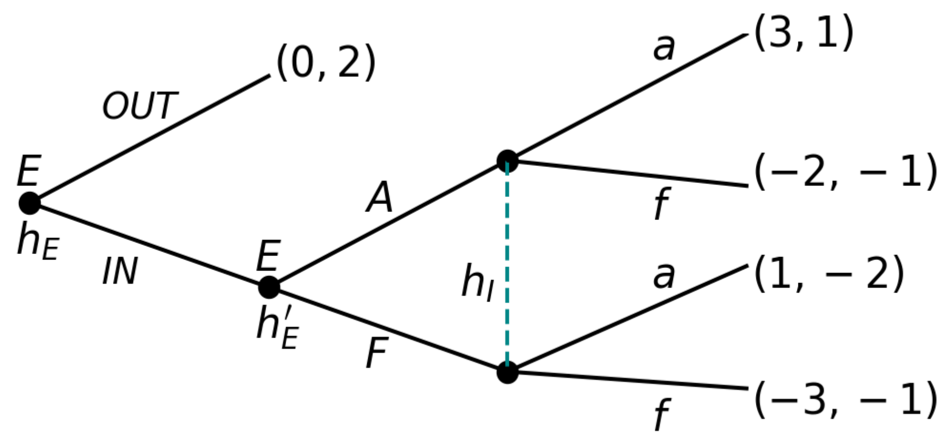


Figure 1

- Pure-strategy NE:
  - $((OUT, F), f)$,
  - $((OUT, A), f)$,
  - $((IN, A), a)$.
- Subgame Perfect: $((IN, A), a)$.
- A peculiar WPBE. $((OUT, A), f)$; $\mu_2((IN, F)|h_I) = 1$.

- Normal form.

	a	**f**
OUT-A	0,2	0,2
OUT-F	0,2	0,2
IN-A	3,1	-2,-2
IN-F	1,-2	-3,-1

- Reduced Normal form.

	a	**f**
OUT-	0,2	0,2
IN-A	3,1	-2,-2

- The Entrant has two information sets, say h_E and h'_E, both singletons. The Incumbent has a single information set, h_I.
- WPBE (OUT-A,f)

$$\begin{aligned} \sigma_E(h_E) &= (1,0);\ (\text{OUT, IN}) \\ \sigma_E(h'_E) &= (1,0);\ (\text{A, F}) \\ \sigma_I(h_I) &= (0,1);\ (\text{a, f}) \\ \mu_I((IN, A)|h_I) &= 0 \end{aligned}$$

This is WPBE but is not SPE; strategies do not specify an equilibrium in the subgme.

Example: unintuitive WPBE

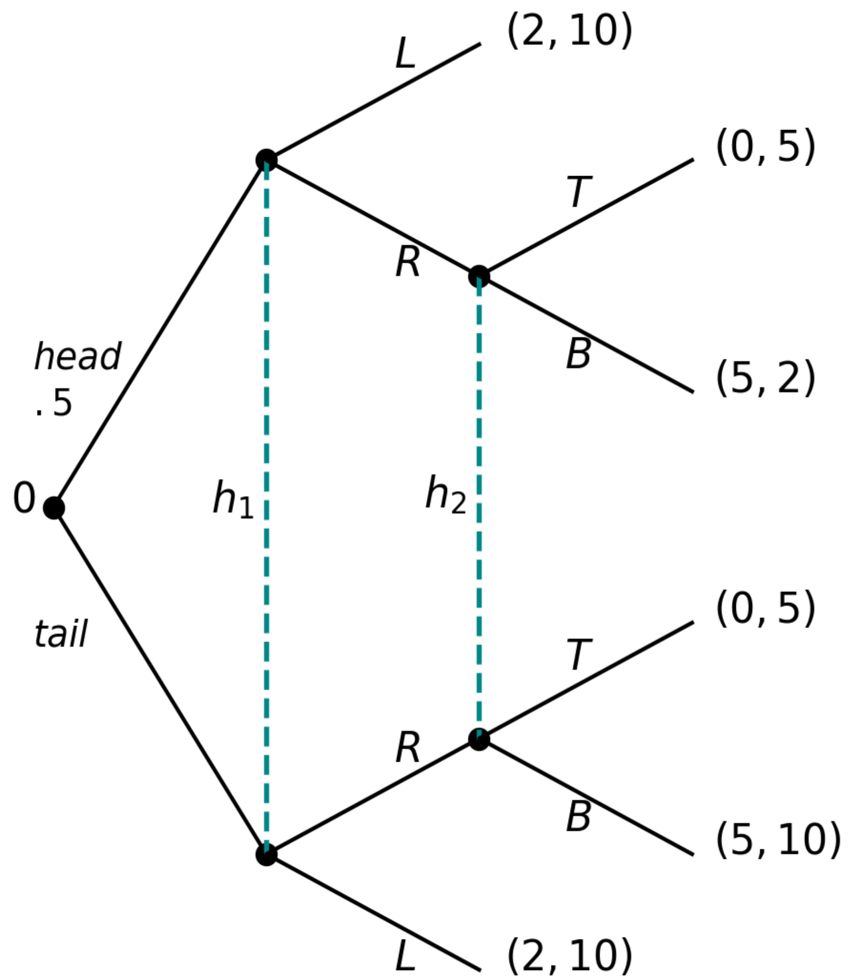


Figure 2

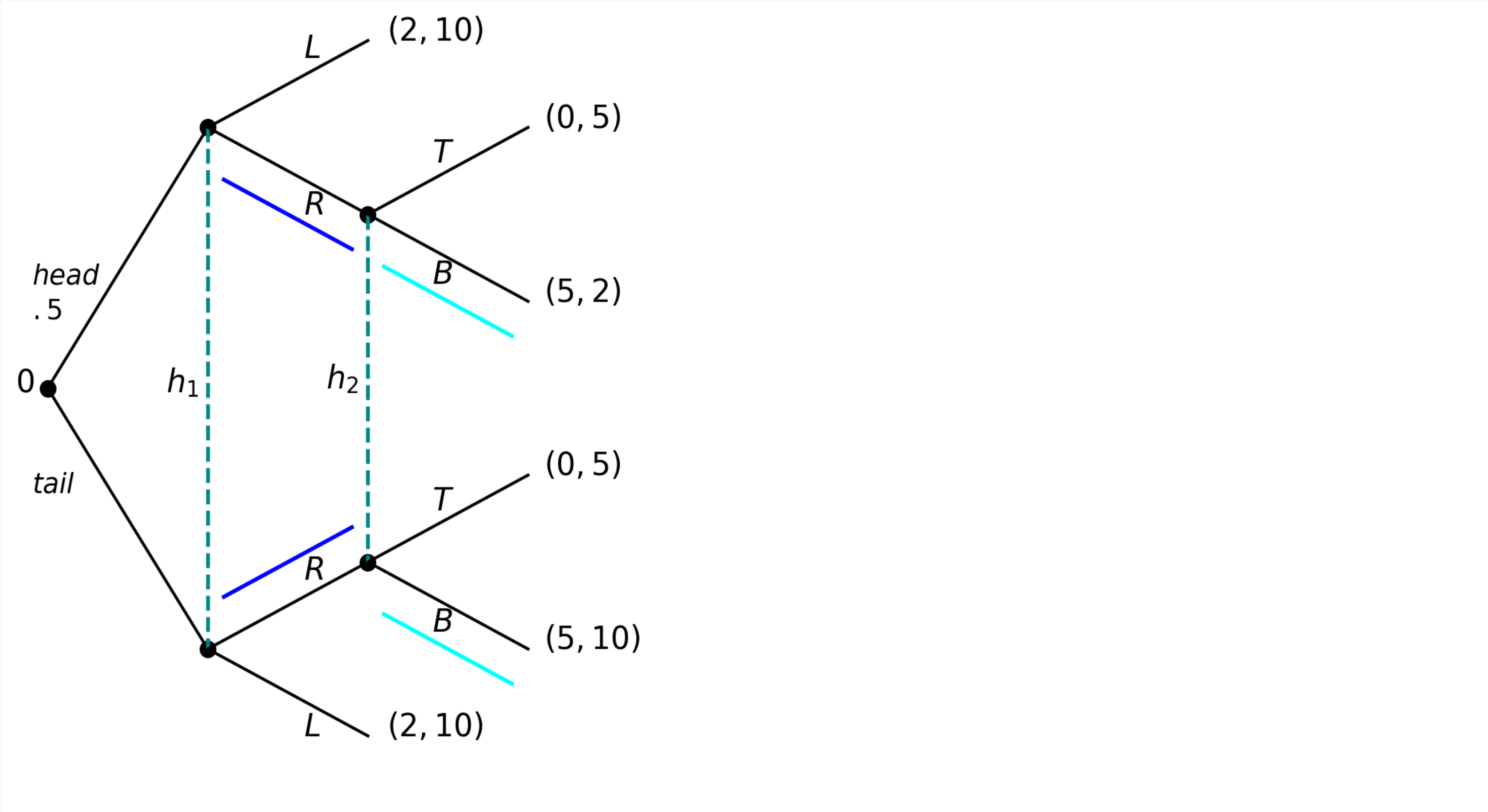


Figure 3

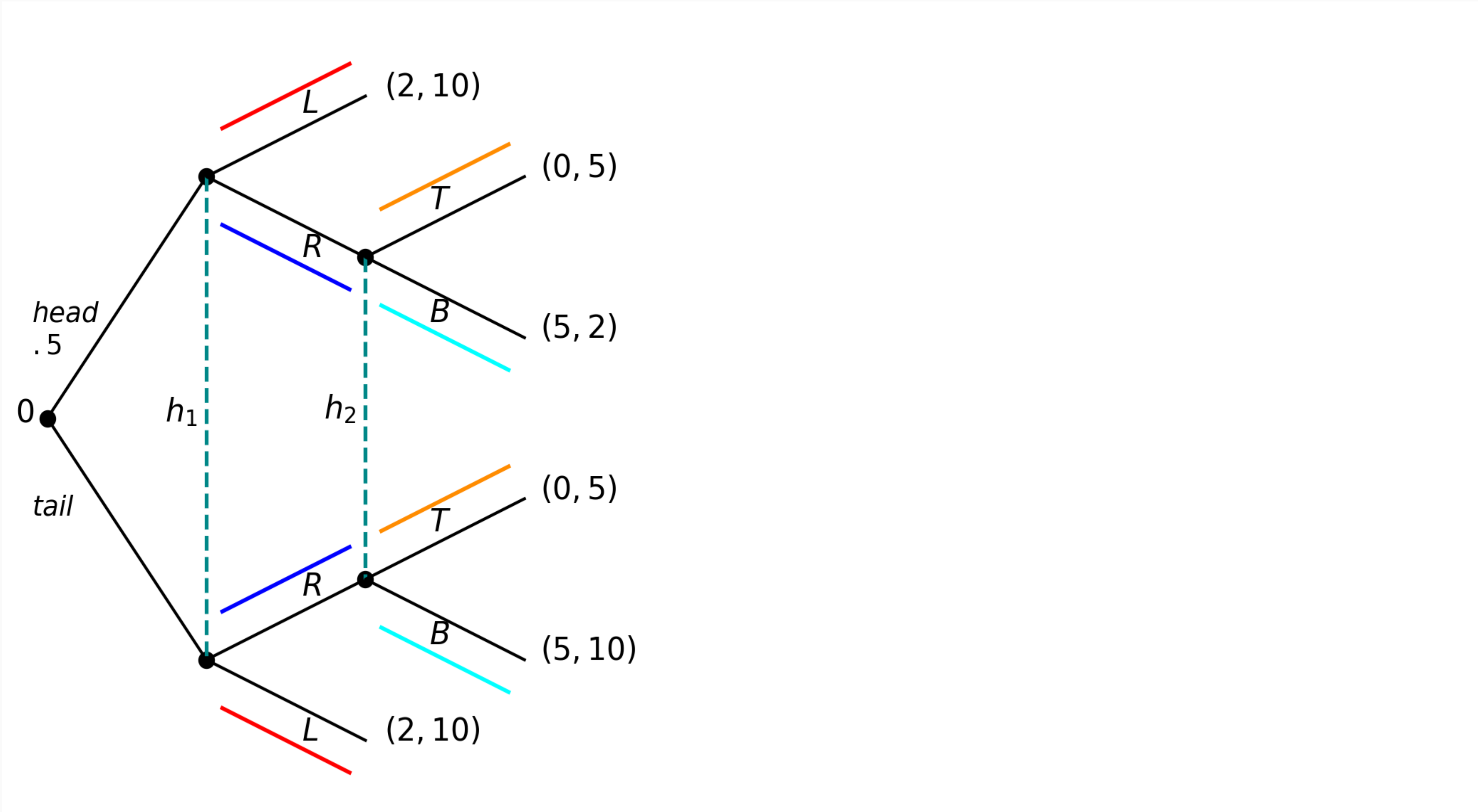


Figure 4

First WPBE ($blue$)

$$\sigma_1(h_1) = (0, 1);\ (L, R). \quad \mu_1(hd|h_1) = \frac{1}{2}$$

$$\sigma_2(h_2) = (0, 1);\ (T, B). \quad \mu_2(hd, R|h_2) = \frac{1}{2}$$

Second WPBE (red)

$$\sigma_1(h_1) = (1, 0);\ (L, R). \quad \mu_1(hd|h_1) = \frac{1}{2}$$

$$\sigma_2(h_2) = (1, 0);\ (T, B). \quad \mu_2(hd, R|h_2) = \frac{9}{10}$$

Remarks

- Many WPBE because it imposes no restrictions on beliefs off the equilibrium path.
- Including beliefs in the definition of equilibrium possibilitates further refinements.
- Red equilibrium and the common prior assumption.
- Finding intuitive properties with examples, define solution concept. Weak as method.

References

Maschler, Michael, Eilon Solan, and Shmuel Zamir. 2013. *Game Theory*. Translated by Ziv Hellman. 1st ed. Cambridge University Press.